

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)		Ductile: Can be drawn into wires or can be easily deformed [plastically] (1) Don't accept reference to malleability Steel: addition of [carbon] atoms inhibits movement / correct reference to impact of atoms on movement (1) of [edge] dislocations (1) Don't accept any reference to planes of atoms slipping over each other – needs movement of dislocation	3			3		
	(b)	(i)	$\varepsilon = 7.5 \times 10^{-4}$ or $7.5 \times 10^{-2}\%$ - no unit penalty		1		1	1	
		(ii)	$\sigma = \varepsilon \times E$ $= 200 \times 10^9 \times 7.5 \times 10^{-4}$ (1) (substitution) (ecf on ε) $= 1.5 \times 10^8$ [N m ⁻² or Pa] (1)	1	1		2	2	
		(iii)	Force = $\sigma \times \text{CSA}$ $= 1.5 \times 10^8 \times \pi (0.3 \times 10^{-3})^2$ (1) (conversion and substitution) (ecf on σ) $= 42[.4 \text{ N}]$ (1) N.B. no marks for using diameter but ecf for powers of 10 error in radius e.g. using 0.3^2		2		2	2	
		(iv)	$W = \frac{1}{2} F \Delta l$ $= \frac{1}{2} \times 42.4 \times 1.8 \times 10^{-3}$ (1) (ecf on force) $= 0.038$ [J] or 0.04 [J] (1) Alternative: Use of $\frac{1}{2} kx^2$ substitution (1) and answer (1)	1	1		2	2	

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	(c)	(i)	T on lower surface between the pillars and C on upper surface - accept in pillars	1			1		
		(ii)	Pre-stressed bar drawn horizontally inside concrete beam from inner edges of supports, towards lower surface (on or below mid-point). Don't accept rectangular shapes which overlap the middle of the concrete beam or go underneath.	1			1		
		(iii)	Concrete / beam under compression (accept reduces tension) / cracks in concrete / beam forced to close (1) Inhibits crack propagation (1)		2		2		
			Question 1 total	7	7	0	14	7	0